



UNIVERSITY OF THE FREE STATE

DEPARTMENT OF COMPUTER SCIENCE AND INFORMATICS

BLOEMFONTEIN/QWAQWA CAMPUS

**SEMESTER TEST 2 MEMO
CSIS2634 & CSIQ2634**

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MODERATOR: Dr J.E. Kotzé

TIME: 2 hours
DATE:
MARKS: 52

Instructions

1. Answer all the questions.
2. Number your answers clearly.
3. When drawing ERDs you must use UML notation. **No Crow's foot notation will be marked.**
4. This is a closed book test. You may not use any notes, text books or electronic devices during the test.

Case Study

Use the following case study to answer the questions that follow.

We are currently in the midst of a Covid-19 pandemic. All shops in South Africa have been directed to keep a roster of shoppers that enter the store. Upon entering the store, the shopper must have their temperature taken. They must also indicate whether they are experiencing any of

the listed symptoms and whether they have recently been in contact with someone who is ill. You have been asked to assist with computerising the roster system.

Question 1

The following roster of data is given:

ShopName	ShopBranch	Shop Branch Code	Shopper Name	Shopper Surname	Shopper IdNumber	Date	Time	Shopper Temperature	Temperature Range	Action
Pick n Pay	Towers	TWR	Paul	Jones	8416548	2020/06/01	10:00	36.2	Normal	No action
Pick n Pay	Towers	TWR	Sally	Jones	4788787	2020/06/01	10:00	33.1	Low	Report
Checkers	Fleurdal	CFL	Jason	Neuman	7848197	2020/06/02	15:43	36.0	Normal	No action
Pick n Pay	Royces	RYC	Paul	Jones	8416548	2020/06/02	11:24	39.2	Elevated	Swab
Checkers	Fleurdal	CFL	Nthabeleng	Moekena	1231867	2020/06/03	9:31	36.2	Low	Report

Normalise the given table to 3NF. Show all your steps. You do not have to draw dependency diagrams, but you must show all dependencies in text format at these steps.

[21]

1NF

Step 1: Eliminate the repeating groups [✓]

There are no repeating groups (no nulls)

Step 2: Identify the primary key [✓]

ShopBranchCode, ShopperIdNumber, Date, Time [✓][✓]

Step 3: Identify all dependencies [✓]

Partial dependencies

ShopBranchCode->ShopBranch, ShopName [✓]

ShopperIdNumber->ShopperName, ShopperSurname [✓]

Transitive dependencies

TemperatureRange->Action [✓]

2NF

Step 1: Write each key component on a separate line [✓]

ShopBranchCode

ShopperIdNumber

Date

Time

ShopBranchCode, ShopperIdNumber, Date, Time

[✓][✓]

Step 2: Assign corresponding dependencies [✓]

Branch(ShopBranchCode, ShopBranch, ShopName) [✓]

Shopper(ShopperIdNumber, ShopperName, ShopperSurname) [✓]

Date

Time

ShopperScan(ShopBranchCode, ShopperIdNumber, Date, Time, ShopperTemperature, TemperatureRange, Action) [✓]

3NF

Step 1: Identify each new determinant [✓]

TemperatureRange [✓]

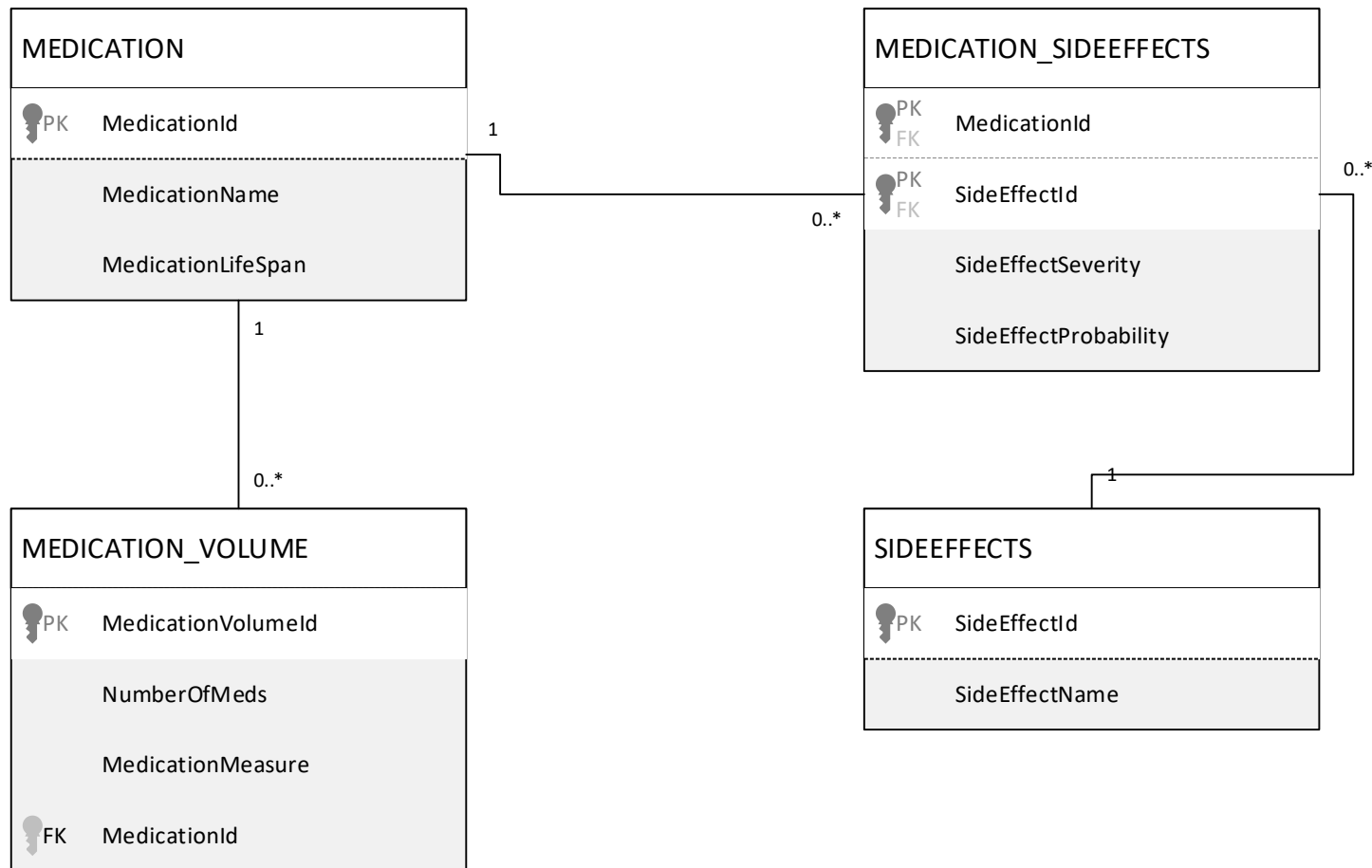
Step 2: Identify the dependent attributes [✓]

TemperatureRange->Action[✓]

Step 3: Remove the dependent attributes from transitive dependencies [✓]

ShopperScan(ShopBranchCode, ShopperIdNumber, Date, Time, ShopperTermperature, TemperatureRange) [✓]

Use the following ERD to answer the question.



Question 2

Write a SQL query to supply a list showing medication name and life span of all available medications. Order the list alphabetically.

[4]

```
select MedicationName,  
       MedicationLifeSpan [✓]  
from Medication [✓]  
order by MedicationName [✓][✓]
```

Question 3

New regulations stipulate that no medication can have more than 100 units (NumberOfMeds). Write a SQL query that will list the name of the medication and the VolumeId for all medication volumes that have more than 100 units.

[5]

```
select MedicationName,  
       MedicationVolumeId [✓]  
from Medication [✓]  
join Medication_Volume [✓]  
on Medication.MedicationId = Medication_Volume.MedicationId [✓]  
and NumberOfMeds > 100 [✓]
```

Question 4

Write a SQL query showing the medication name, side effect name, probability and severity for all medications that have a side effect with a probability of 70 or more and a severity of 3 or more.

[8]

```
select MedicationName,  
       SideEffectName,  
       SideEffectProbability,  
       SideEffectSeverity [✓]  
from Medication [✓]
```

```

join Medication_SideEffects [✓]
  on Medication.MedicationId = Medication_SideEffects.MedicationId [✓]
join SideEffects [✓]
  on SideEffects.SideEffectId = Medication_SideEffects.SideEffectId [✓]
and SideEffectSeverity >= 3 [✓]
and SideEffectProbability >= 70 [✓]

```

Question 5

Write a SQL query to generate a list of all medications (show medication name and id) as well as the number of volume packages that each medication has.

[7]

```

select MedicationName,
       MedicationId, [✓]
       count(*) [✓]
from Medication [✓]
  join Medication_Volume [✓]
    on Medication.MedicationId = Medication_Volume.MedicationId [✓]
group by MedicationName,
         MedicationId [✓][✓]

```

Question 6

Write a SQL query listing all medications that have no side effects at all.

[7]

```

select MedicationName [✓]

```

```
from Medication [✓]  
left join Medication_SideEffects [✓][✓]  
    on Medication.MedicationId = Medication_SideEffects.MedicationId [✓]  
where Medication_SideEffects.SideEffectId is null [✓][✓]
```